Hrid

MEMBERSHIP

Stars in catalog	666
with Gaia GSP-Phot	399 (60%)
with radial velocity	29
Stream id (sID)	71

GEOMETRY · great-circle fit

Centroid RA, Dec	279.7, 40.6 deg
Centroid l, b	70.1, 19.3 deg
Pole RA, Dec	188.2, 2.2 deg
Length (φ1, 2–98%)	64.8 deg
Width σ(φ2), robust	1.46 deg

KINEMATICS

pmRA* med ± σ	-5.96 ± 1.63 mas/yr
pmDE med ± σ	20.72 ± 3.31 mas/yr
μφ1* med ± σ	20.40 ± 3.44 mas/yr
μφ2 med ± σ	6.93 ± 0.97 mas/yr
Parallax (cat.) med	0.293 ± 0.148 mas
Parallax (wtd, ZP-corr.)	0.304 ± 0.010 mas
Distance	d = 3.3 ± 0.1 kpc
Physical length	3.71 kpc
RV median ± σ(MAD)	-258.0 ± 96.9 km/s
RV gradient dV/dφ1	5.79 km/s/deg

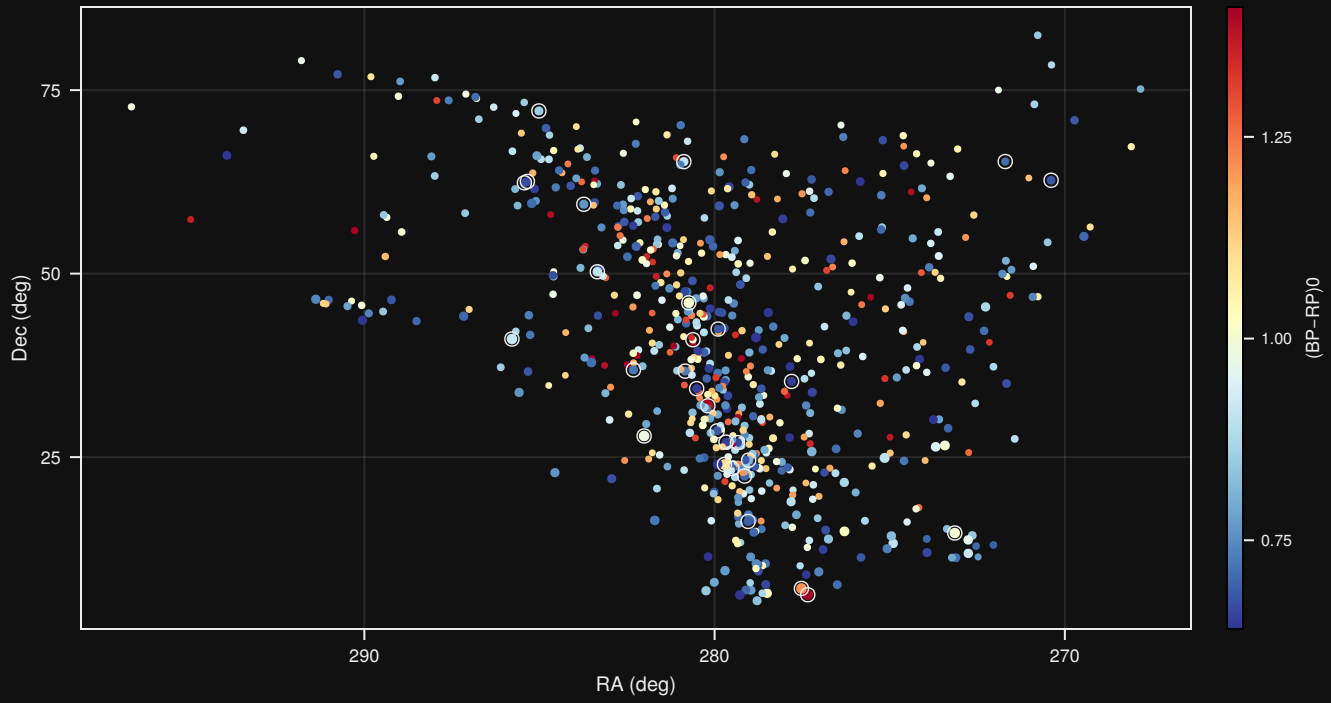
PHOTOMETRY · POPULATIONS

G0 range (median)	10.9–20.0 (18.3)
(BP−RP)0 median	0.90
GSP-Phot [M/H] (n=399)	-1.62 ± 0.46
GSP-Phot Teff median	5442 K
RUWE > 1.4	1.0 %
PM clusters (largest)	41 (33 stars)

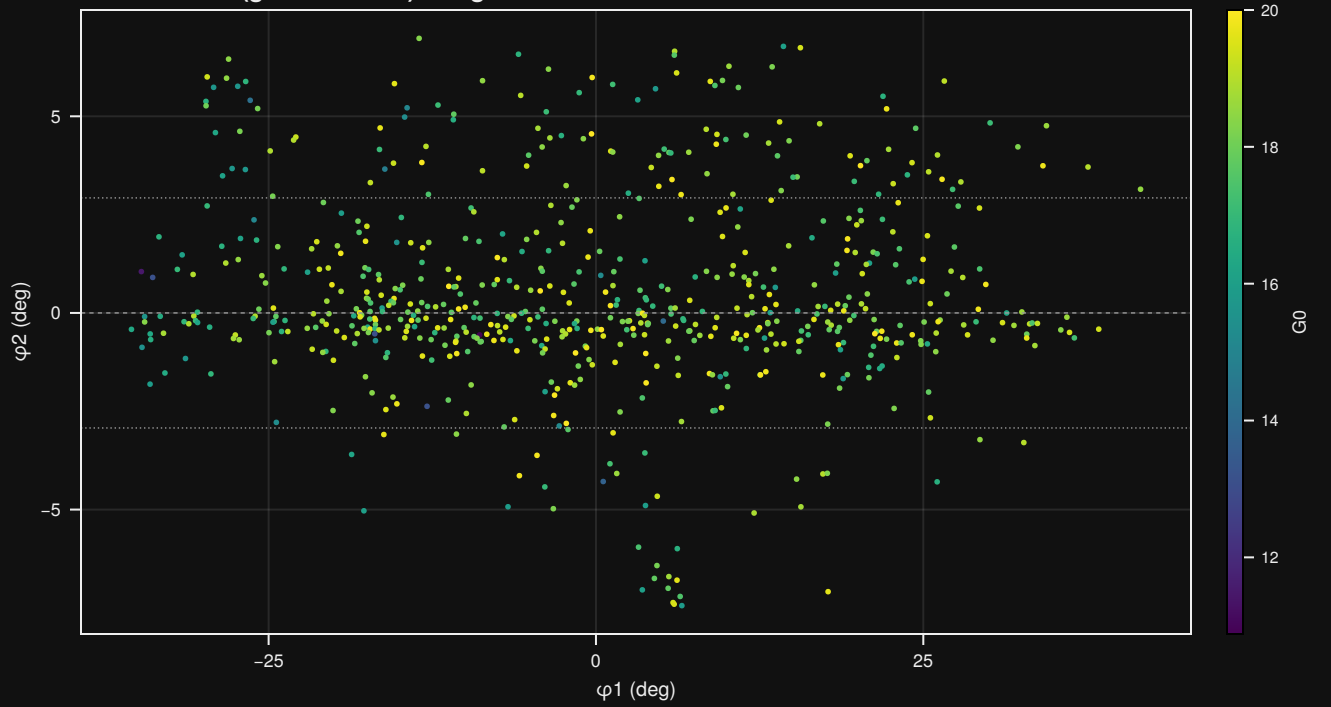
IBATA+2024 · TABLE 3

n catalog | paper	666 | 666
n_RV catalog | paper	29 | 29
<[Fe/H]> paper	-1.13
Stellar mass (paper)	2000 Msun
Pole offset vs paper	0.6 deg

Sky map · catalog members (n = 666) · size ∝ brightness · ring = has RV

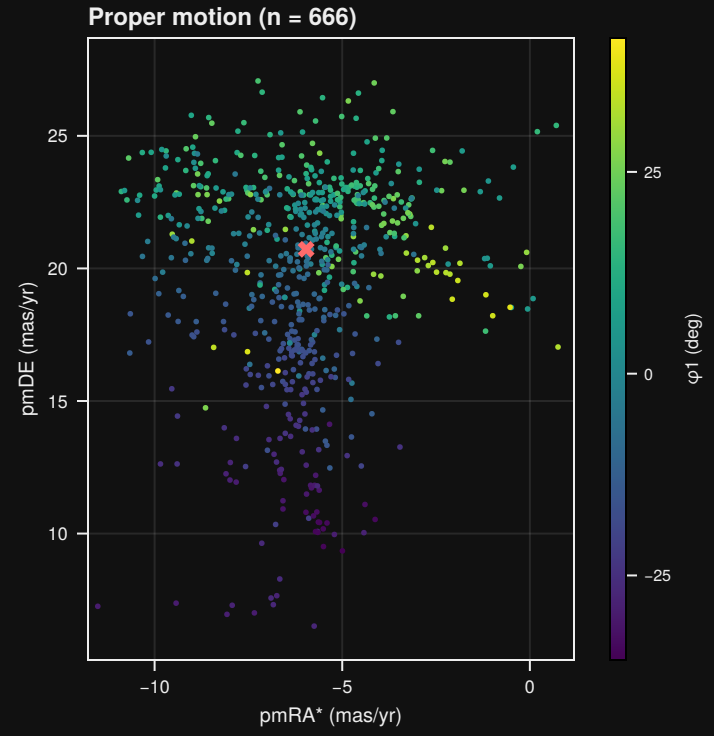


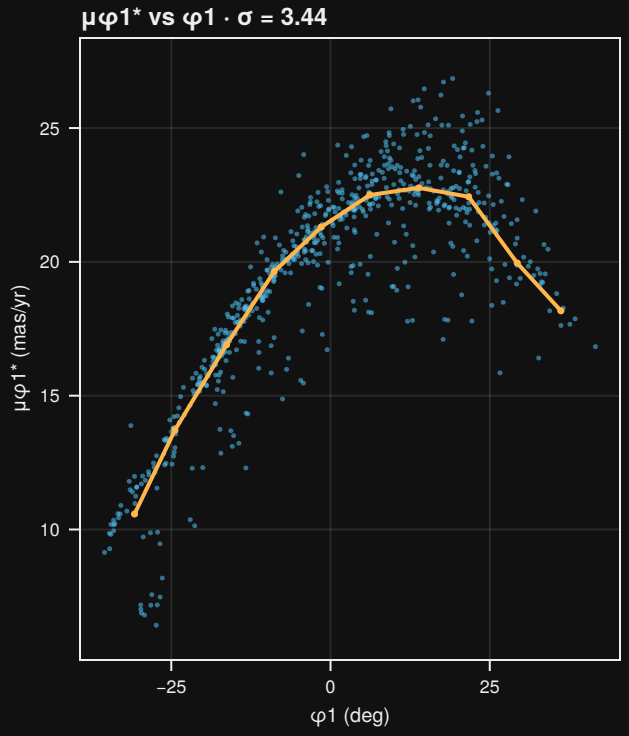
Stream frame (great-circle fit) · length 64.8° · width σ 1.46°

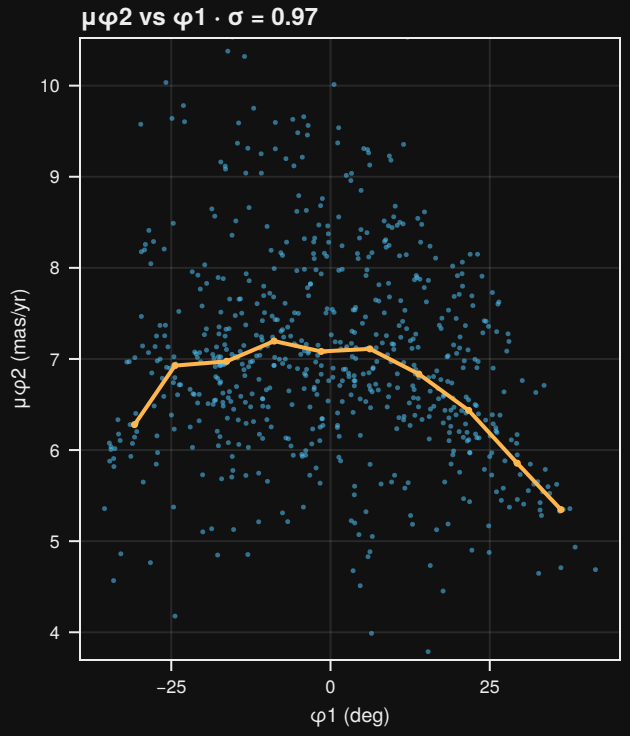


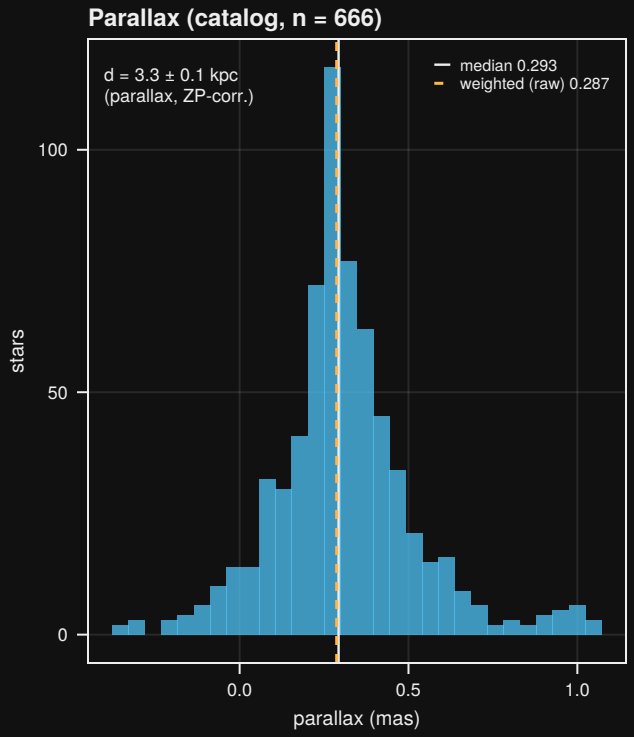
Hrid

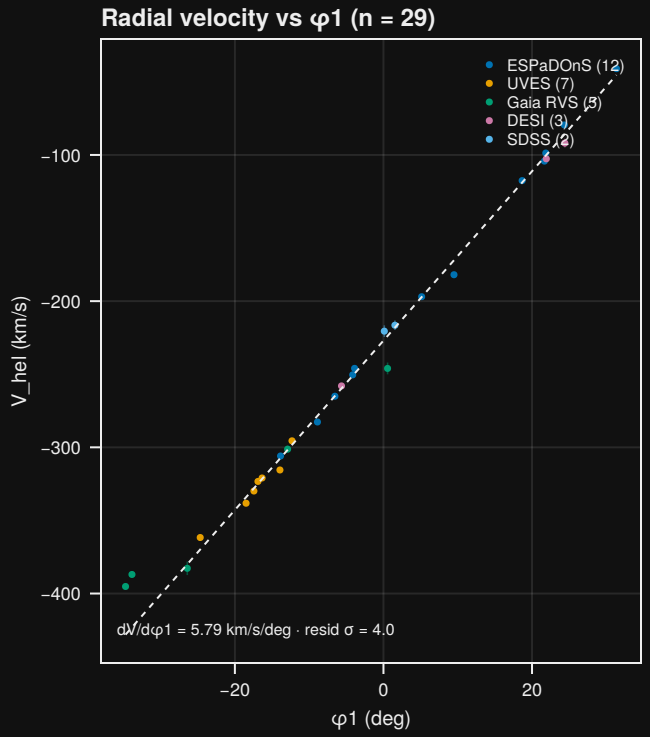
Kinematics · proper motion · parallax · radial velocity

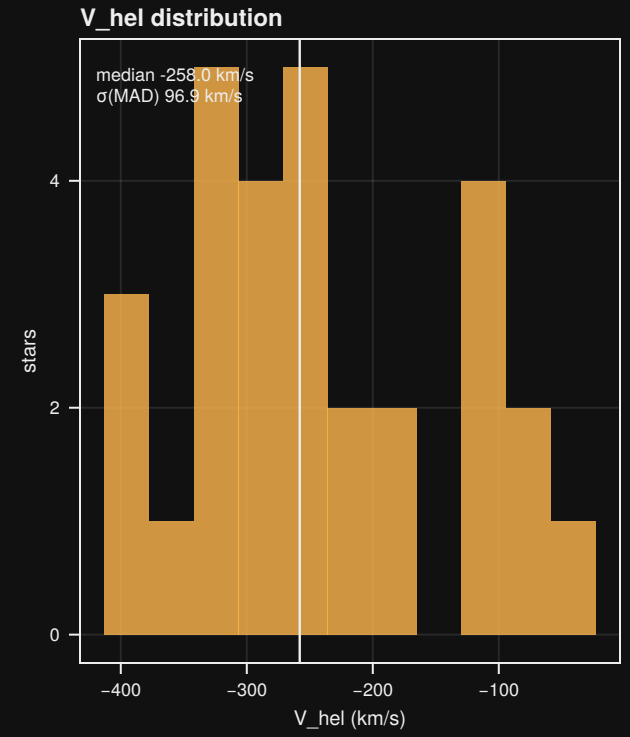






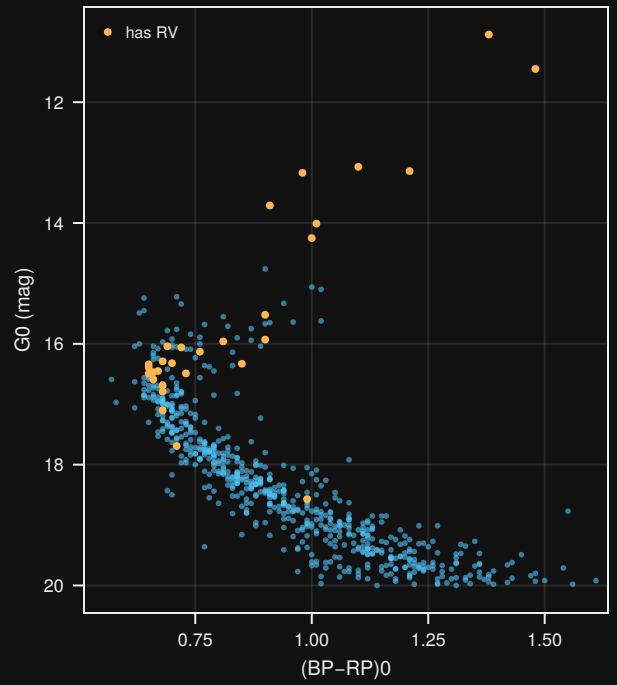




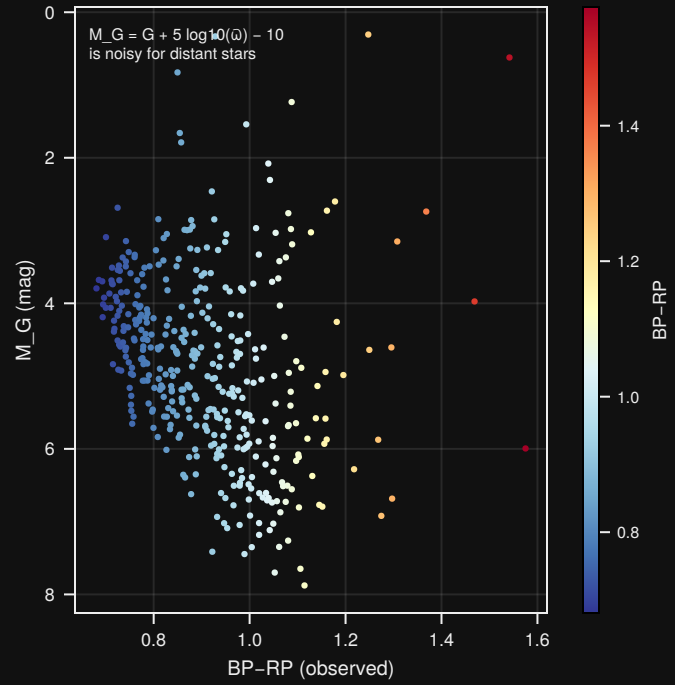


Hrid

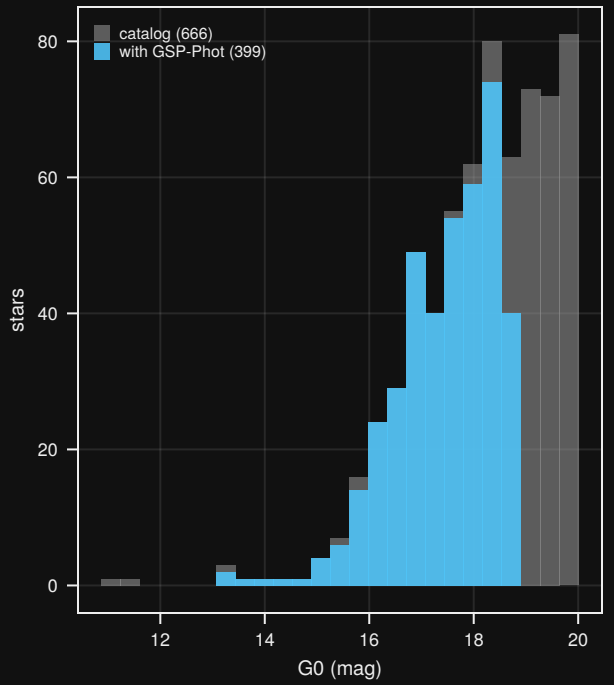
CMD · dereddened (n = 666)



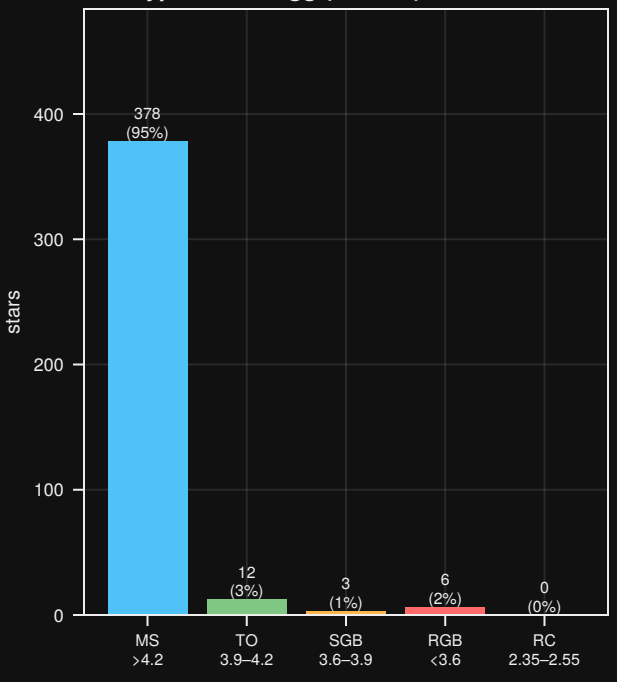
HR diagram · ϖ>0 (n = 393/399)



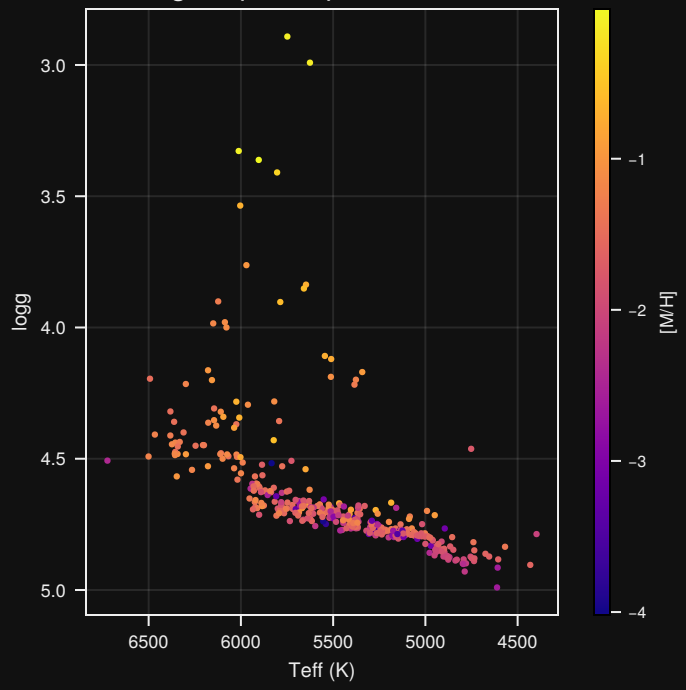
G0 & GSP-Phot coverage



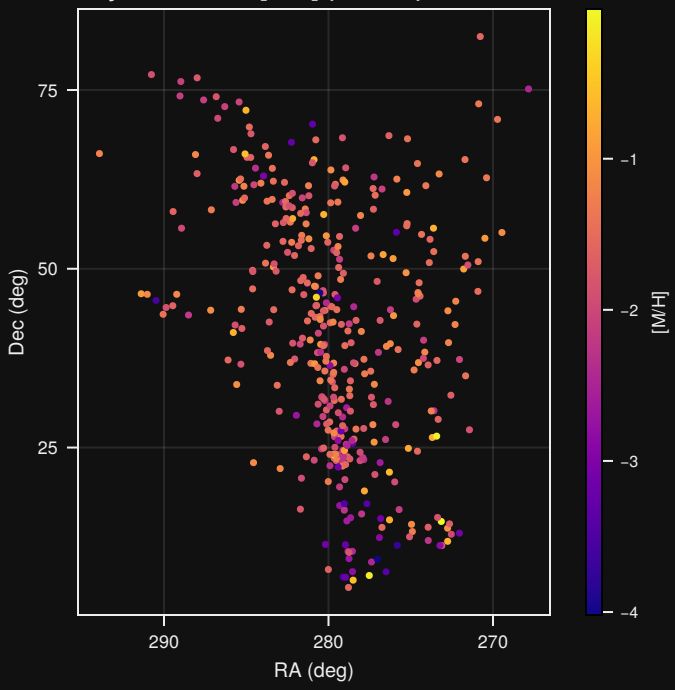
Star types from logg (n = 399)



Kiel diagram (n = 399)



Sky · GSP-Phot [M/H] (n = 399)

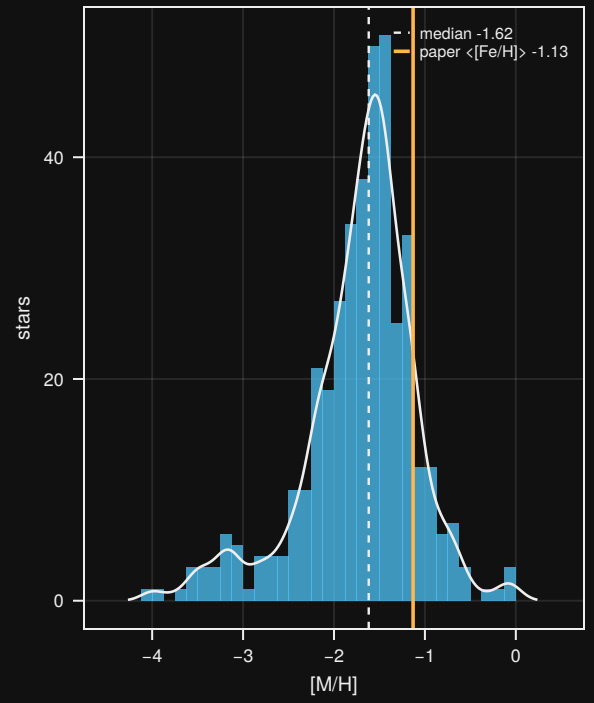


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≳ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

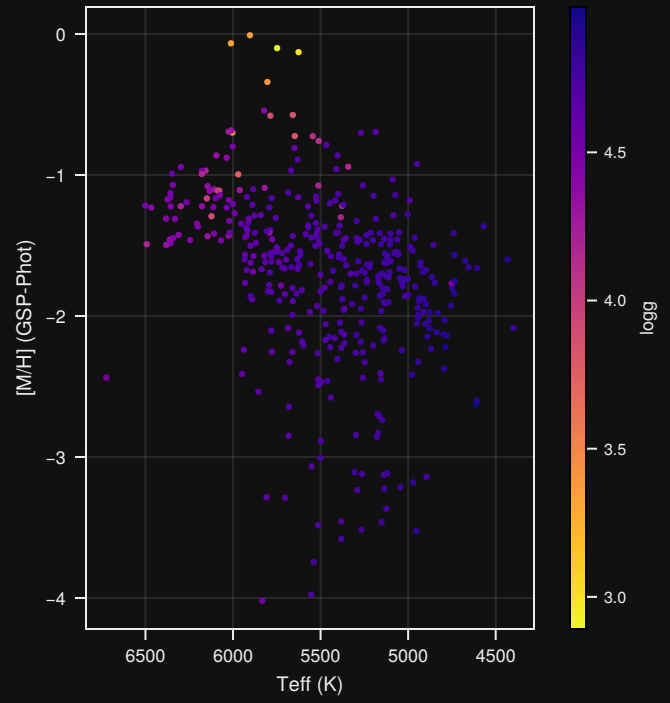
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Metallicity · age · astrometric quality

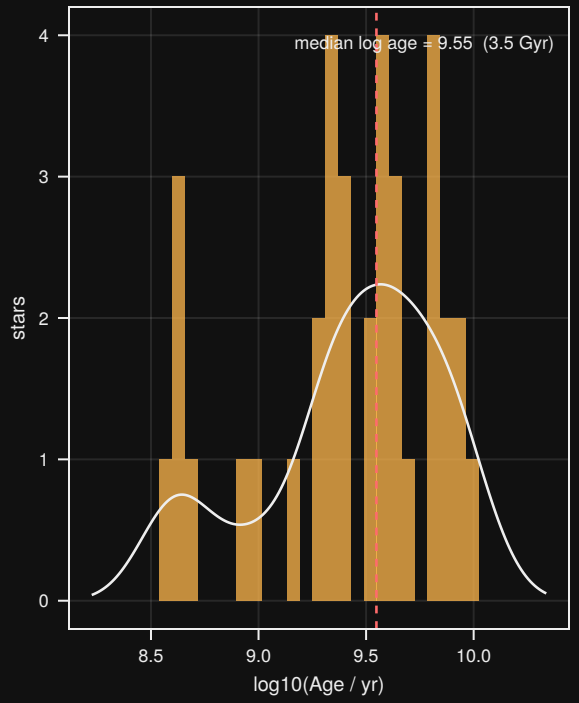
[M/H] GSP-Phot (n = 399)



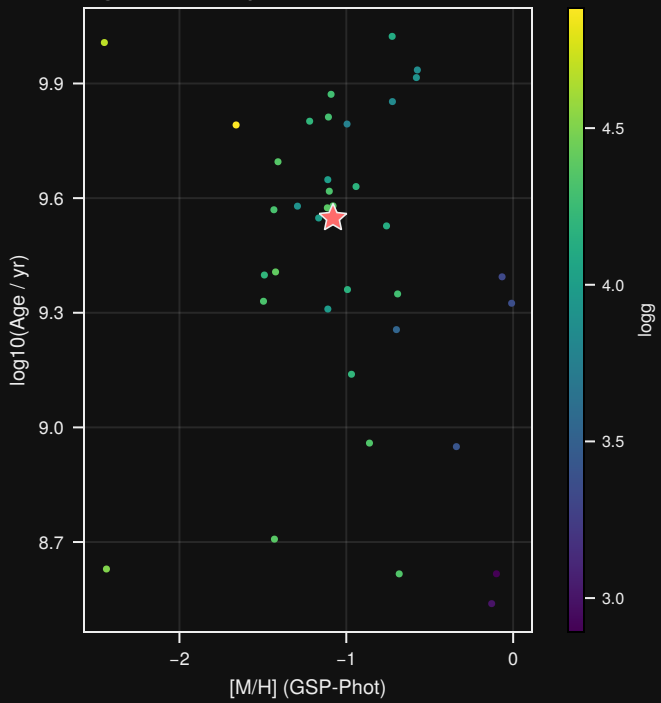
Teff vs [M/H] (n = 399)



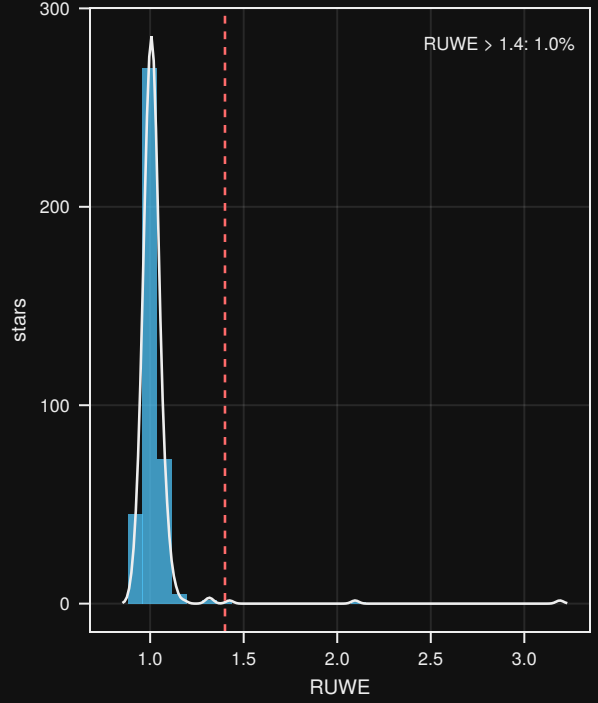
FLAME age (n = 37)



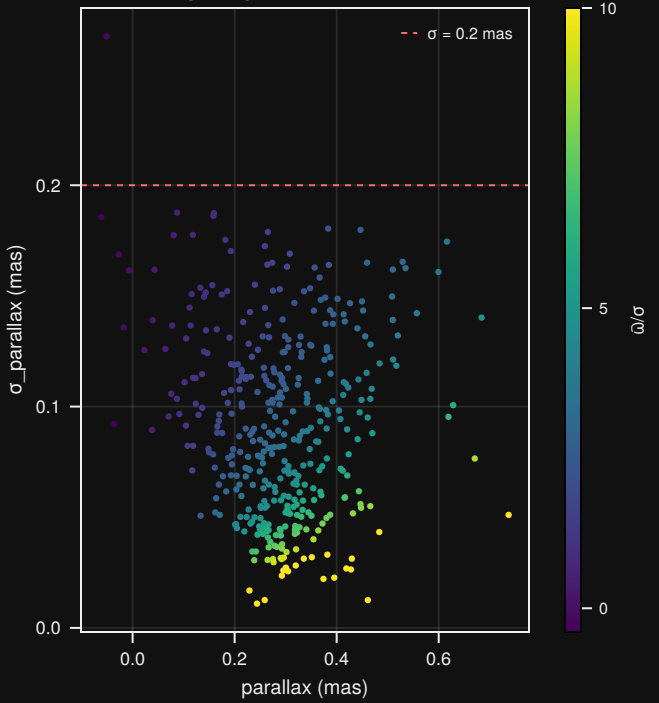
Age–metallicity (n = 37)



RUWE (n = 398)

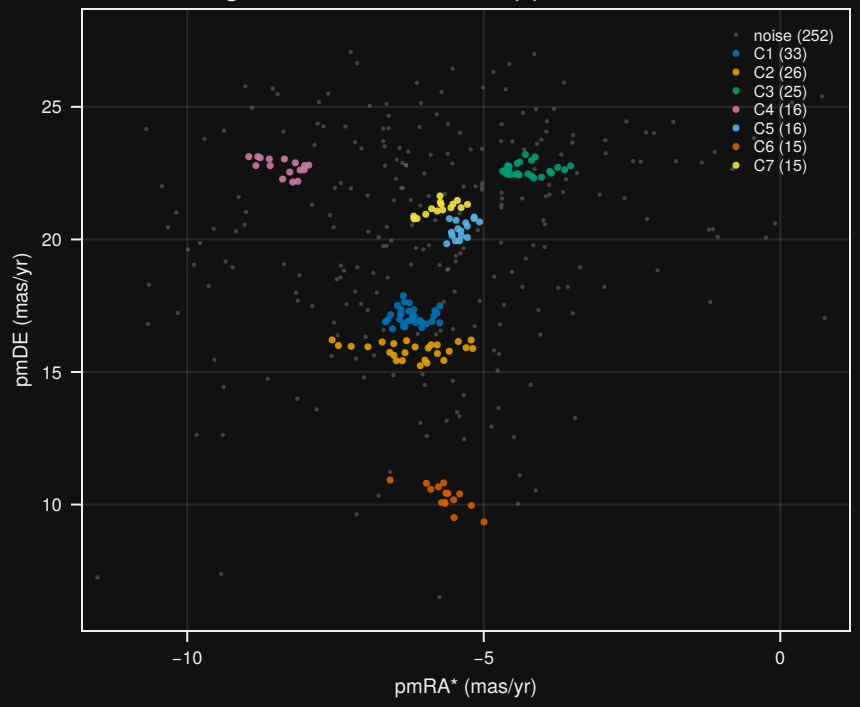


Parallax quality (n = 399)

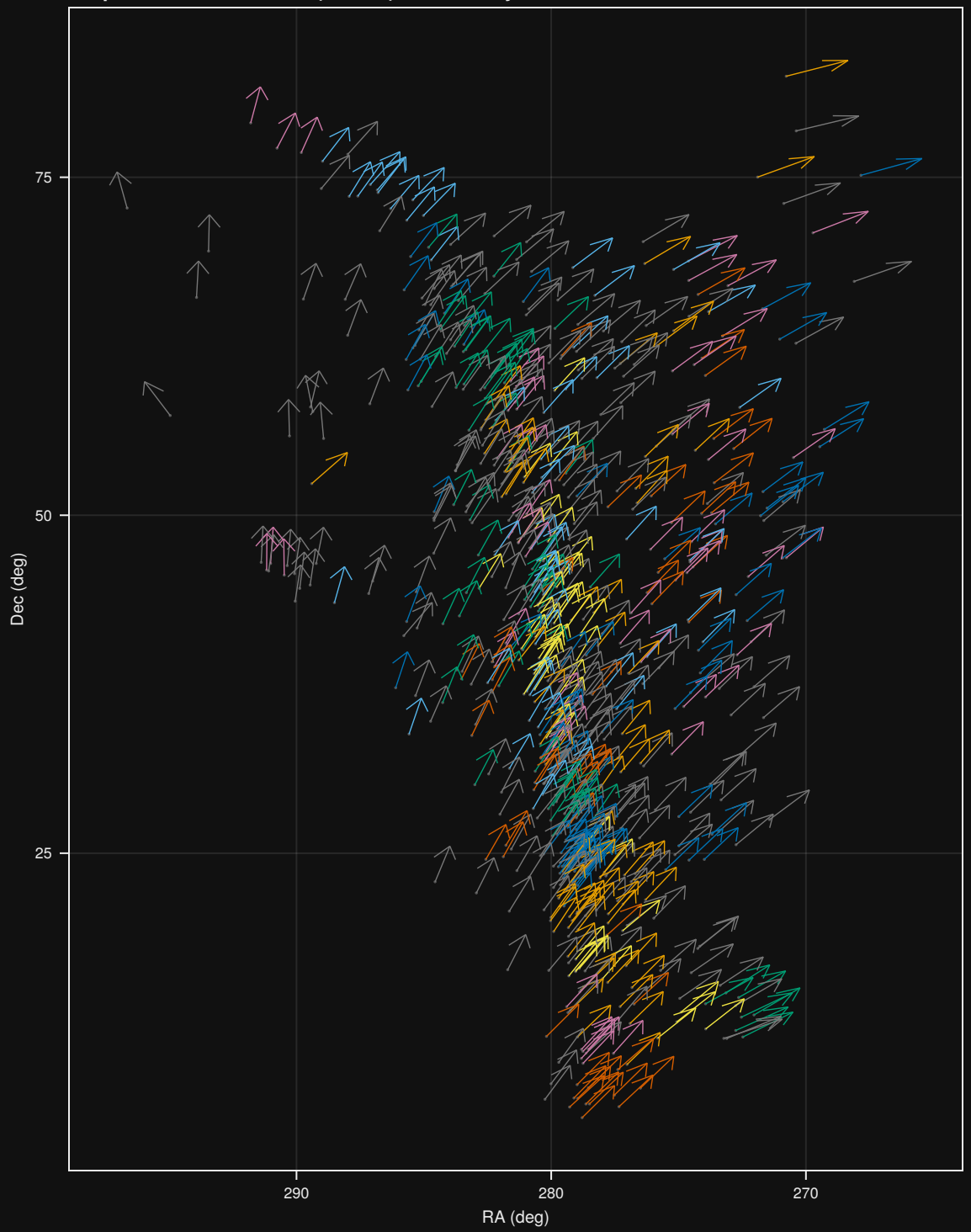


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PM clustering · HDBSCAN · 41 cluster(s) · noise 38%



Proper-motion directions (arrows) coloured by cluster



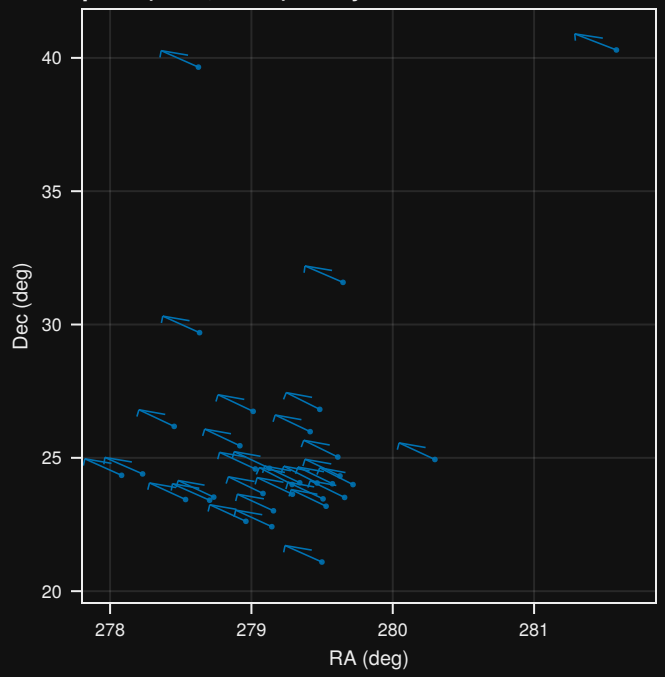
Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	252	-5.80	20.66	2.06	-1.63	8
C1	33	-6.26	17.02	0.21	-1.69	4
C2	26	-6.11	15.91	0.60	-1.91	0
C3	25	-4.39	22.58	0.30	-1.56	1
C4	16	-8.32	22.78	0.43	-1.46	0
C5	16	-5.40	20.36	0.18	-1.49	2
C6	15	-5.65	10.40	0.22	-2.47	2
C7	15	-5.73	21.20	0.32	-1.49	0
C8	13	-9.96	22.84	0.74	-1.40	1

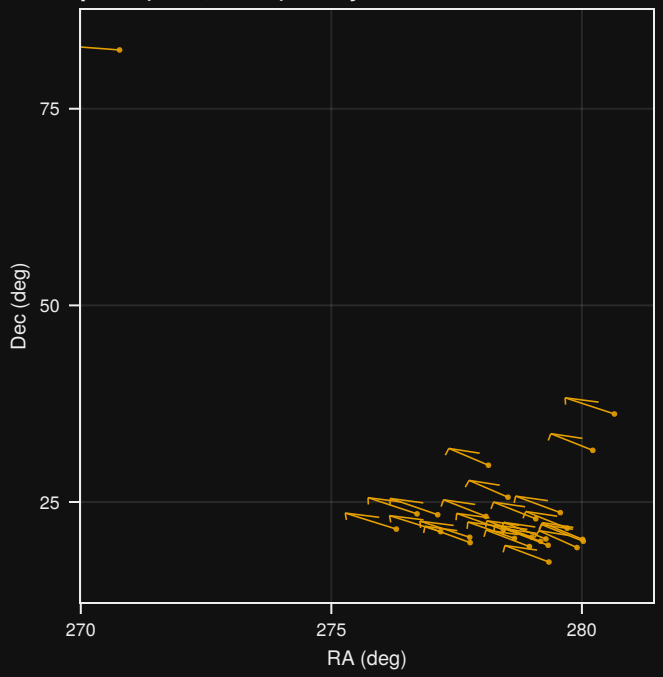
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Per-cluster proper-motion vector maps

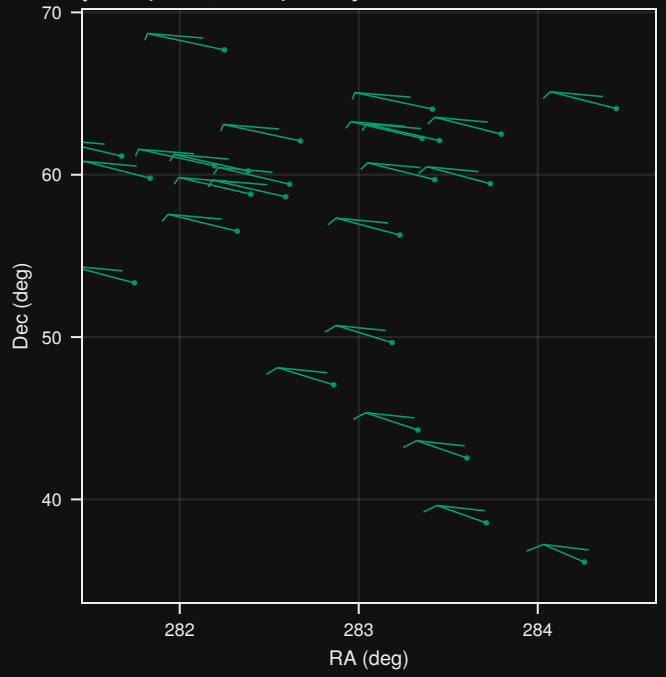
Cluster C1 · n = 33
pm = (-6.26, 17.02) mas/yr



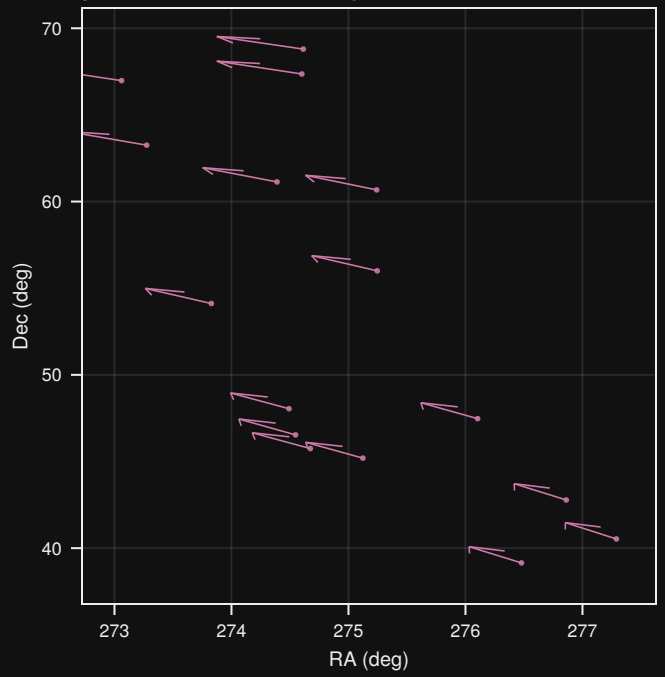
Cluster C2 · n = 26
pm = (-6.11, 15.91) mas/yr



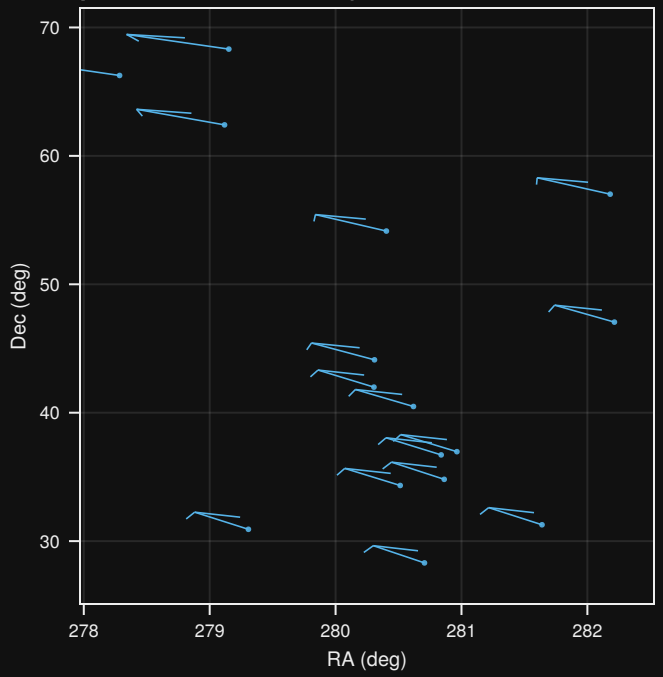
Cluster C3 · n = 25
pm = (-4.39, 22.58) mas/yr



Cluster C4 · n = 16
pm = (-8.32, 22.78) mas/yr



Cluster C5 · n = 16
pm = (-5.40, 20.36) mas/yr



Noise · n = 252

